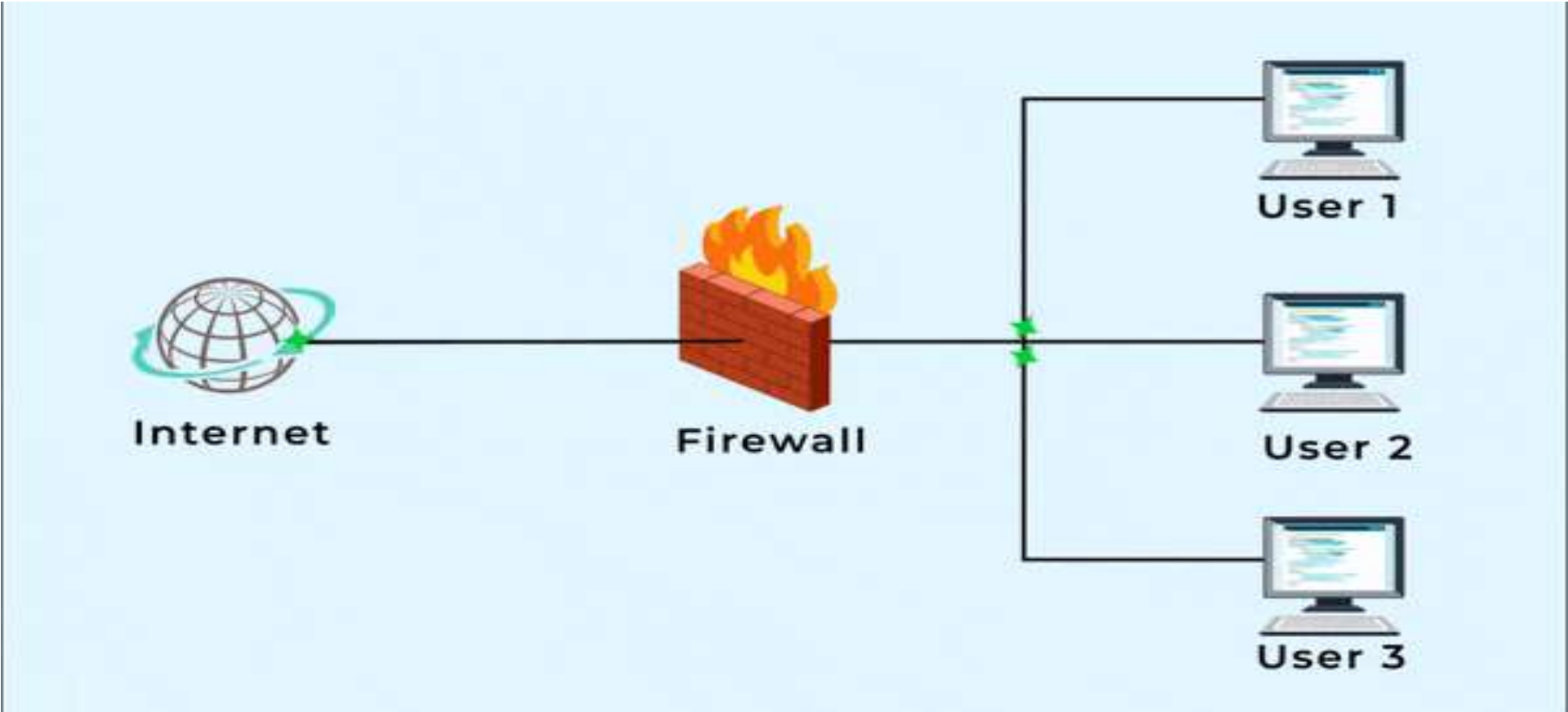
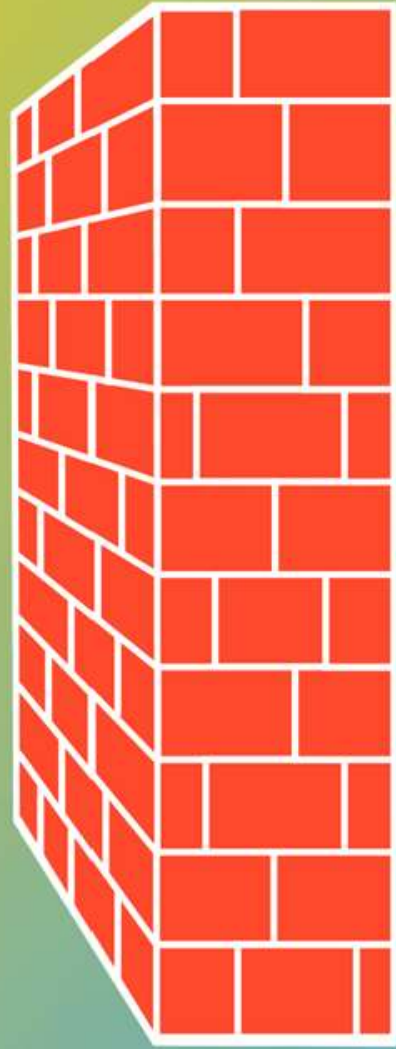


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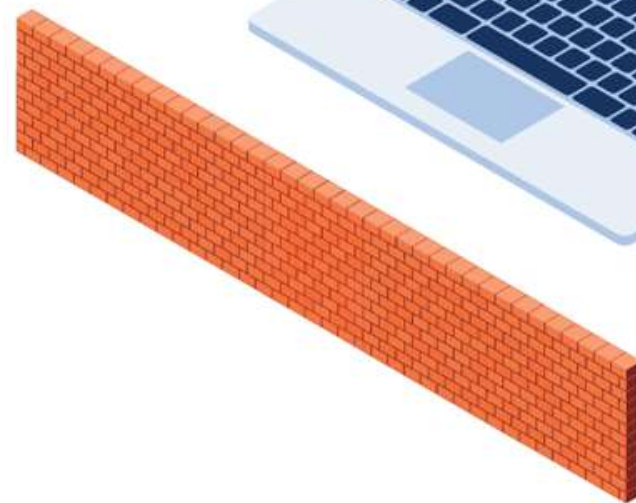
Firewall & Intrusion







Firewall





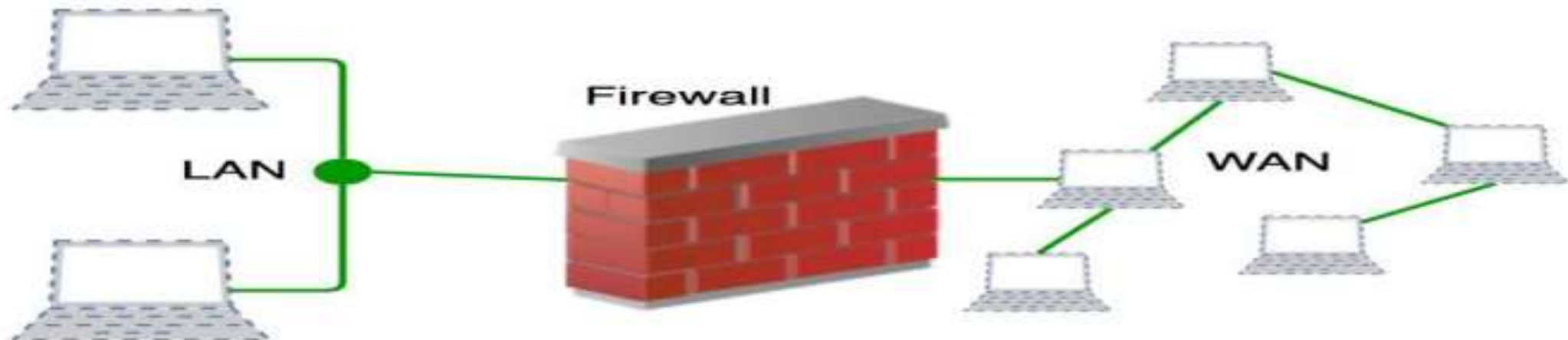
Unit V	Firewall And Intrusion	(08 Hours)
Introduction, Computer Intrusions. Firewall Introduction, Characteristics and types, Benefits and limitations. Firewall architecture, Trusted Systems, Access Control. Intrusion detection, IDS: Need, Methods, Types of IDS, Password Management, Limitations and Challenges.		

Introduction of Firewall

- A firewall is a network **security device, either hardware or software-based,**
- which monitors all incoming and outgoing traffic and
- based on a defined set of security rules it **accepts, rejects or drops that specific traffic.**
- **Accept** : allow the traffic.
- **Reject** : block the traffic but reply with an “unreachable error”
- **Drop** : block the traffic with no reply

Other Definition:

Firewalls act as a barrier between your internal network and external networks, such as the Internet. They allow or block traffic based on factors like the protocol, port, source IP address, destination IP address, and more.



How Does a Firewall Work?

Packet filtering:-

traffic is examined at the network protocol layer and allowed or blocked based on source/destination IP addresses, protocols, and ports numbers.

Stateful inspection:-

monitor connections by remembering the **status of network connections**.
To differentiate legitimate packets for existing connections from suspicious traffic.

Application-level gateway (Proxy server):-

verifies contents at the application layer. This provides deep inspection capabilities.

Next generation firewall (NGFW):-

NGFWs combine packet filtering and application layer filtering including intrusion detection and prevention.

Firewalls use a clearly defined rule set to examine traffic and make allow/deny decisions about access. Anything not explicitly allowed by a firewall rule is blocked.

The firewall rule set is based on factors like:

Source and destination IP addresses

- Network protocols (TCP, UDP, ICMP, etc.)
- Port numbers
- Applications
- User identity
- Geographic location
- Time of day
- And many others

Role of Firewalls

Containing threats and stopping lateral movement between network segments.

- Providing visibility through robust logging and reporting.
- Shielding devices with weak security postures.
- Enabling policy enforcement based on users, groups, applications, and content.

Characteristics of Firewalls

1. Traffic Filtering

As traffic passes through the firewall, each packet is examined and matched to the criteria in the rule set to determine whether it should be allowed or blocked.

2. Access Control

Firewalls govern access between network zones by allowing specific types of traffic to pass while explicitly denying all other traffic.

3. Network Segmentation

Internal firewalls can subdivide large private networks into smaller segments. Network segmentation contains threats and limits the impact of breaches.

4. Security Logging & Monitoring

Logging and reporting enable administrators to monitor activity, analyze trends, and be alerted about suspicious traffic.

5. Stateful Inspection

allowing return traffic for established sessions while dropping other packets that don't match any approved flows.

6. SSL/SSH Inspection

Many firewalls can decrypt outbound encrypted traffic, scan its contents, and then re-encrypt before forwarding permitted connections. Decryption defeats threats trying to conceal malicious payloads within encrypted tunnels.

7. IPS/IDS Integration

to detect behavioral anomalies and known threat signatures. This provides advanced threat defense.

8. User Identity Integration

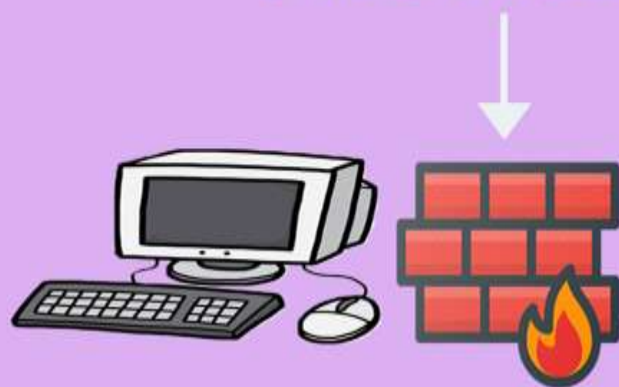
allows firewall rules to be written based on user identity rather than just IP address. This enables policies that follow users regardless of device or network location.

Types of Firewall

Host- based Firewalls(hardware based) :installed on each network node which controls each incoming and outgoing packet.

- It is a software application or part of the operating system.
- Host-based firewalls **are needed because** network firewalls cannot provide protection inside a trusted network.
- Host firewall protects **each host** from attacks and unauthorized access.

Host-based Firewall



Secure Switch



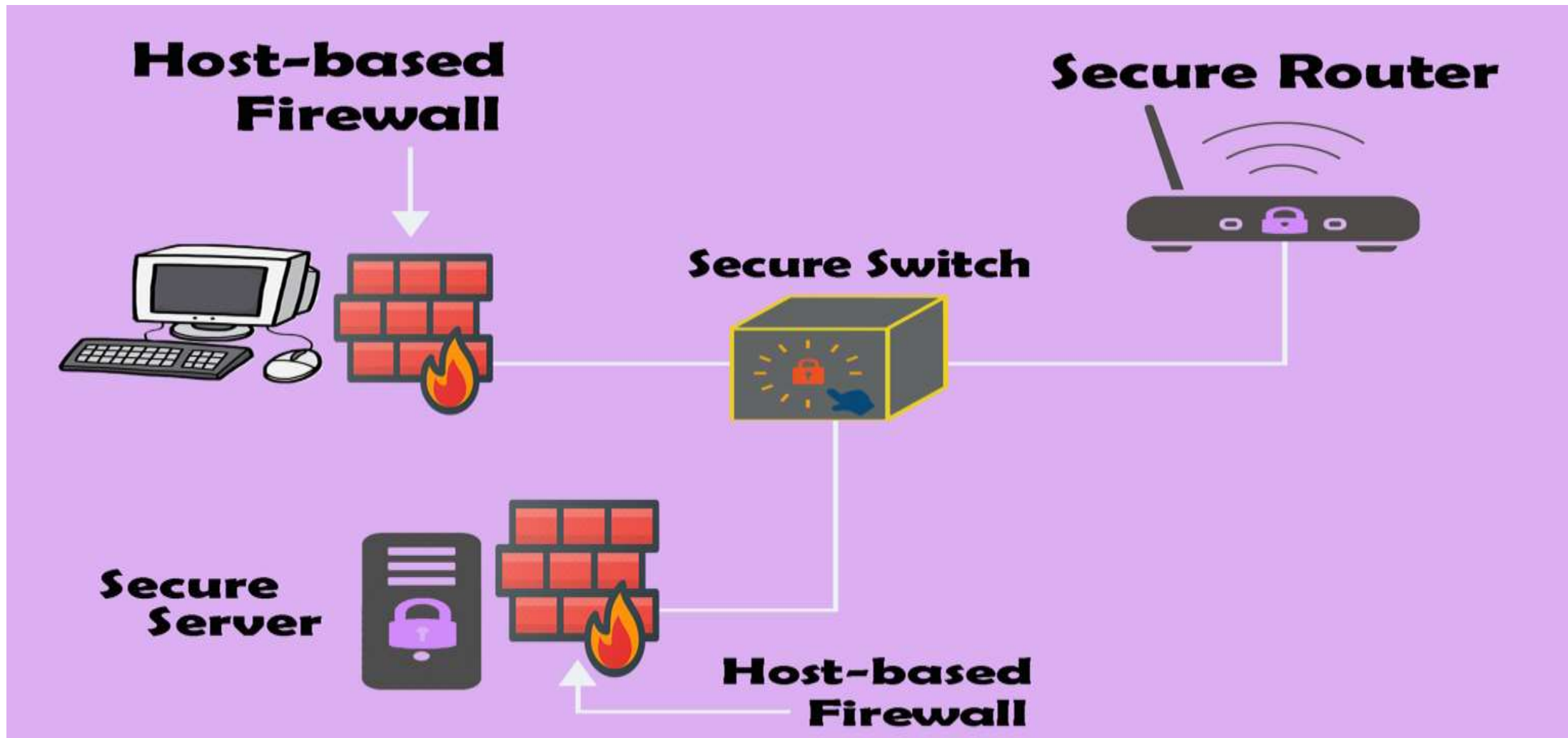
Secure Router



Secure Server



Host-based Firewall



Network-based Firewalls(Software based) :

Network firewall **function on network level.**

- firewalls filter all incoming and outgoing traffic **across the network.**
- It protects the **internal network** by filtering the traffic using rules defined on the firewall.
- A Network firewall might have two or more **network interface cards (NICs).**
- A network-based firewall is usually a dedicated system with proprietary software installed.

Network based FIREWALL

Host based FIREWALL



Legitimate Traffic



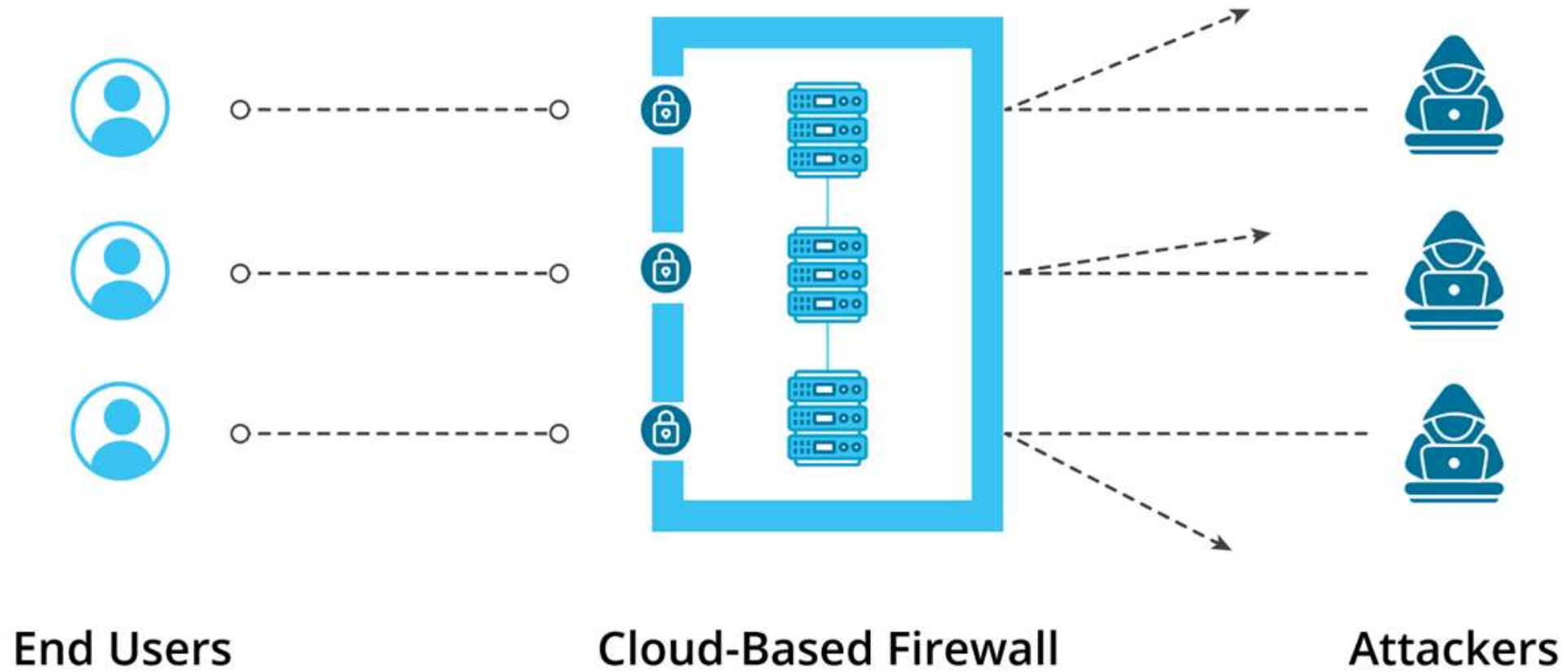
Non-Legitimate Traffic



Cloud Firewalls

- **Amazon Web Services (AWS) security groups** – Virtual firewalls can control traffic in and out of AWS resources like EC2 instances and VPCs.
- **Microsoft Azure firewall** – Azure firewall policies secure virtual networks with stateful packet filtering, intrusion detection, and identity-based controls.
- **Google Cloud firewall** – Cloud firewall rules manage connections between resources like VMs, serverless applications, and services. These integrate natively with other cloud security controls.

Cloud-Based Firewall



Different Types Of Firewalls Explained

Software Firewalls

A software firewall is installed on the host device. Since it is attached to a specific device, it has to utilize its resources to work. Therefore, it is inevitable for it to use up some of the system's RAM and CPU.

Packet-Filtering Firewalls

Packet-Filtering Firewalls serves as an inline security checkpoint attached to a router or switch. As the name suggests, it monitors network traffic by filtering incoming packets according to the information they carry.

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Cloud Firewalls

A cloud firewall or firewall-as-a-service (FaaS) is a cloud solution for network protection. Like other cloud solutions, it is maintained and run on the Internet by third-party vendors.

Secureb4.io

Proxy Firewalls

It serves as an intermediate device between internal and external systems communicating over the Internet. It protects a network by forwarding requests from the original client and masking it as its own.

1

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Hardware Firewalls

Hardware firewalls are security devices that represent a separate piece of hardware placed between an internal and external network (the Internet). This type is also known as an Appliance Firewall.

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Next-Generation Firewalls

The next-generation firewall is a security device that combines a number of functions of other firewalls. It incorporates packet, stateful, and deep packet inspection.



SECUREB4
We Strengthen Your Security

Circuit-Level Gateways

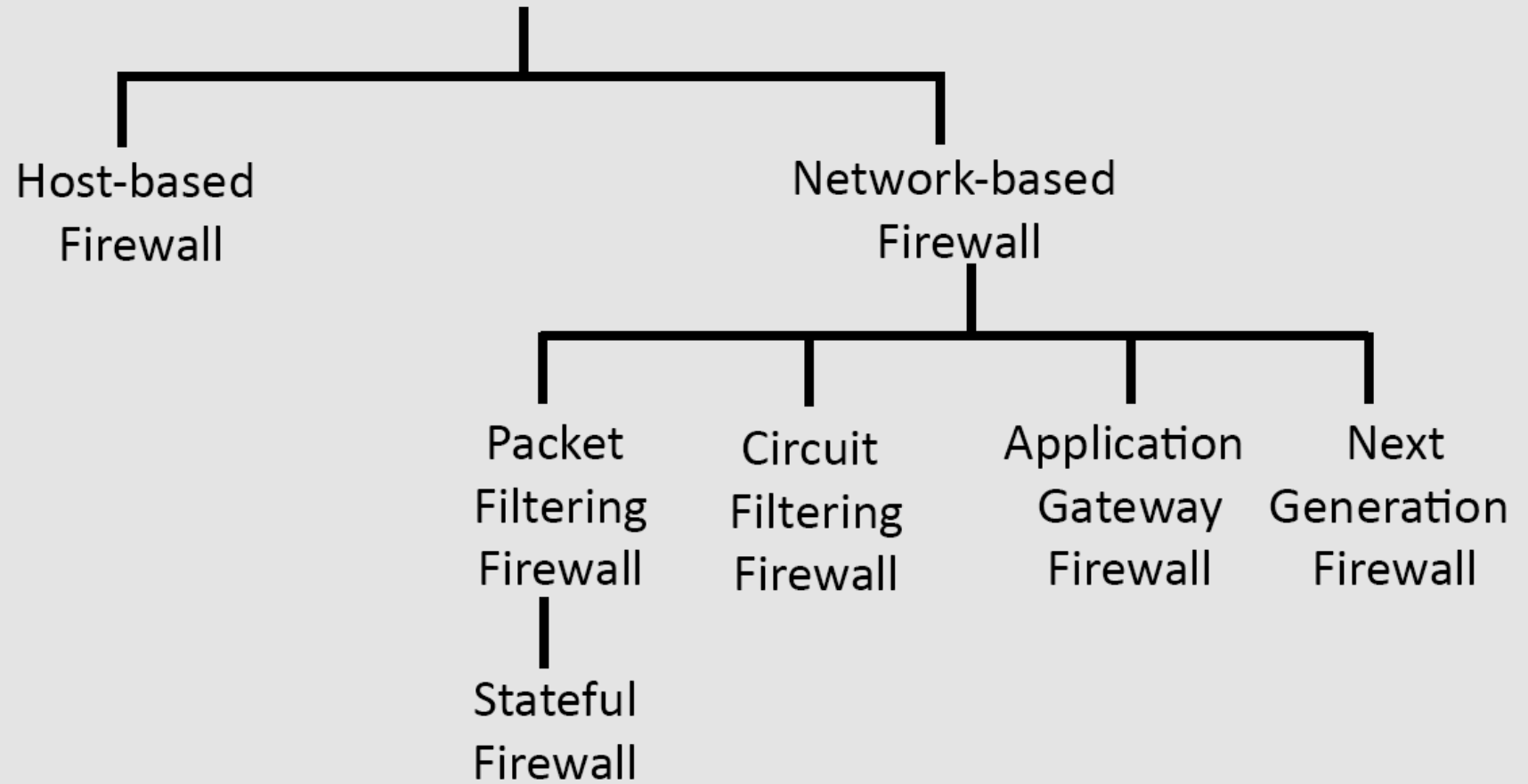
Circuit-level gateways are a type of firewall that work at the session layer of the OSI model, observing TCP (Transmission Control Protocol) connections and sessions.

Stateful Inspection Firewalls

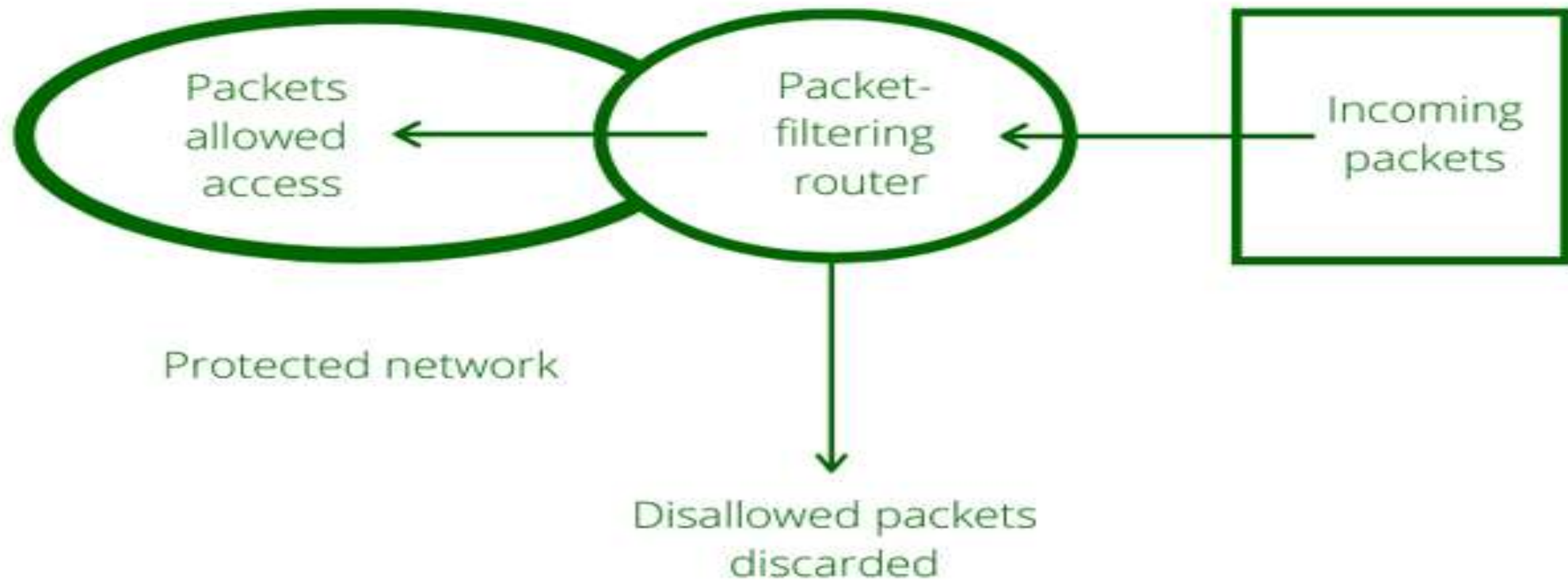
A stateful inspection firewall keeps track of the state of a connection by monitoring the TCP 3-way handshake.

FIREWALL TYPE	ADVANTAGES	DISADVANTAGES
Packet filtering firewall	■ A single device can filter traffic for the entire network ■ Efficient and fast at processing packets ■ Enables complex security policies through filtering on protocol headers ■ Inexpensive ■ Minimal impact on other resources, network performance, end-user experience	■ Incapable of filtering at the application layer ■ Lacks broad context of other firewall options ■ Can be difficult to securely configure ■ Lacks features like user authentication, logging ■ Vulnerable to spoofing attacks ■ Access controls lists can be difficult to set up and manage
Circuit-level gateway	■ Provides privacy for data passing in/out of private network ■ More efficient processing traffic than application-level gateways ■ Relatively inexpensive ■ Easier to set up and manage ■ Minimal impact on end-user experience	■ Protects circuits (network sessions) rather than individual packets ■ Requires modification to network protocol stack ■ Incapable of content filtering ■ Should be used in conjunction with other firewall technologies ■ Does not offer application-layer monitoring
Application-level gateway	■ Capable of detecting and blocking attacks not visible at the OSI model network or transport layers ■ Obscures private network details ■ Protects user anonymity ■ Enables more fine-grained security controls	■ Complex to configure and maintain ■ High processing overhead ■ Requires a proxy be set up for every network application in use ■ Can affect network performance

Firewall



Packet filtering firewall



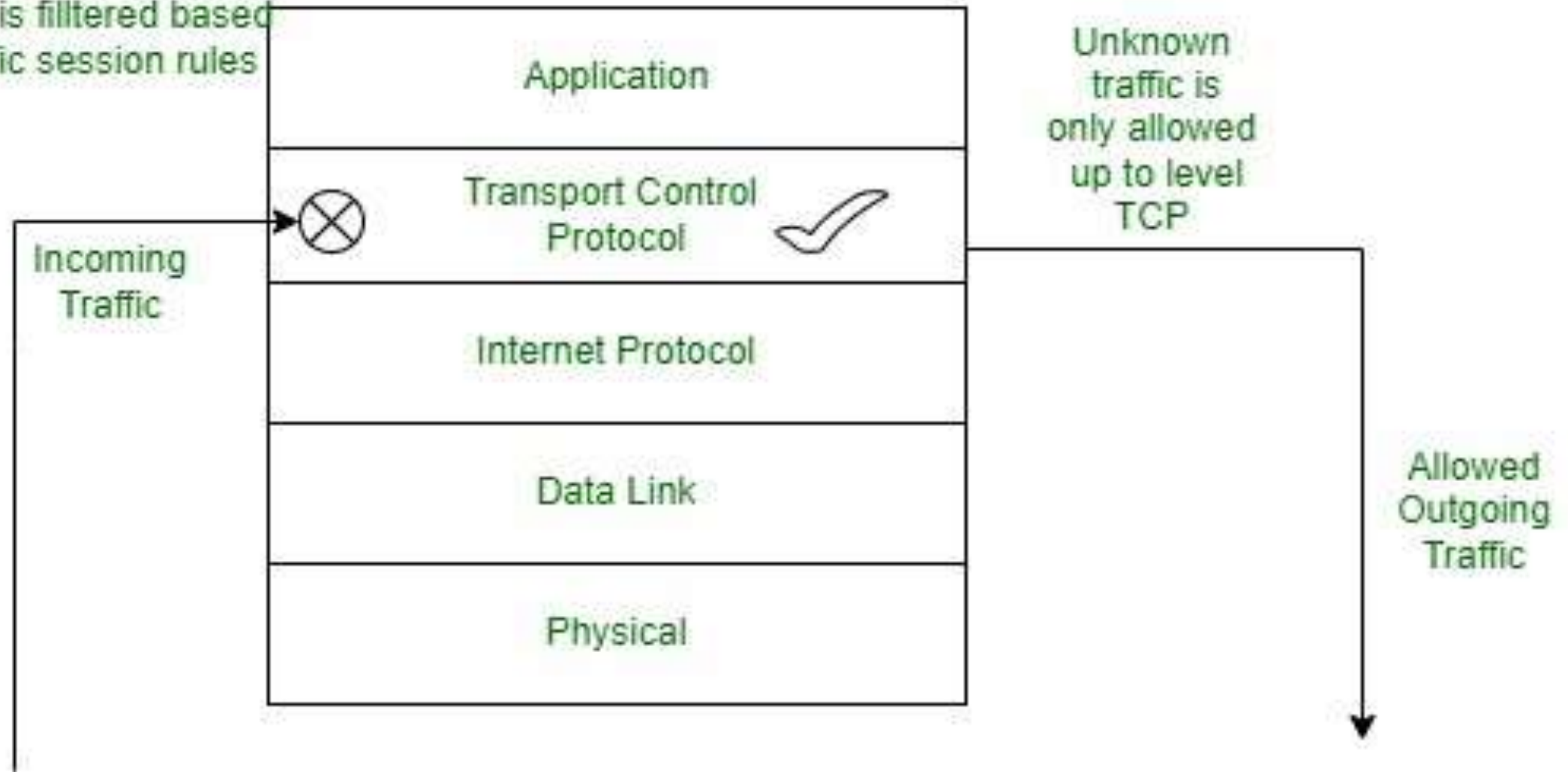
- Packet filters consider only the most basic attributes of each packet, and they don't need to remember anything about the traffic since each packet is examined in isolation. For this reason, they can decide packet flow very quickly.
- Example: Filter can be set to block all UDP segments and all Telnet connections. This type of configuration prevents outsiders from logging onto internal hosts using Telnet and insider from logging onto external hosts using Telnet connections.

Stateful inspection firewall

- State-aware devices examine each packet and keep track of whether that packet is part of an established TCP or other network sessions. Such provision offers more security than packet filtering or [circuit monitoring](#) alone but takes a greater toll on network performance.
- Another variant of stateful inspection is the multilayer inspection firewall, which considers the flow of transactions in process across multiple protocol layers of the seven-layer open systems interconnection (OSI) model.

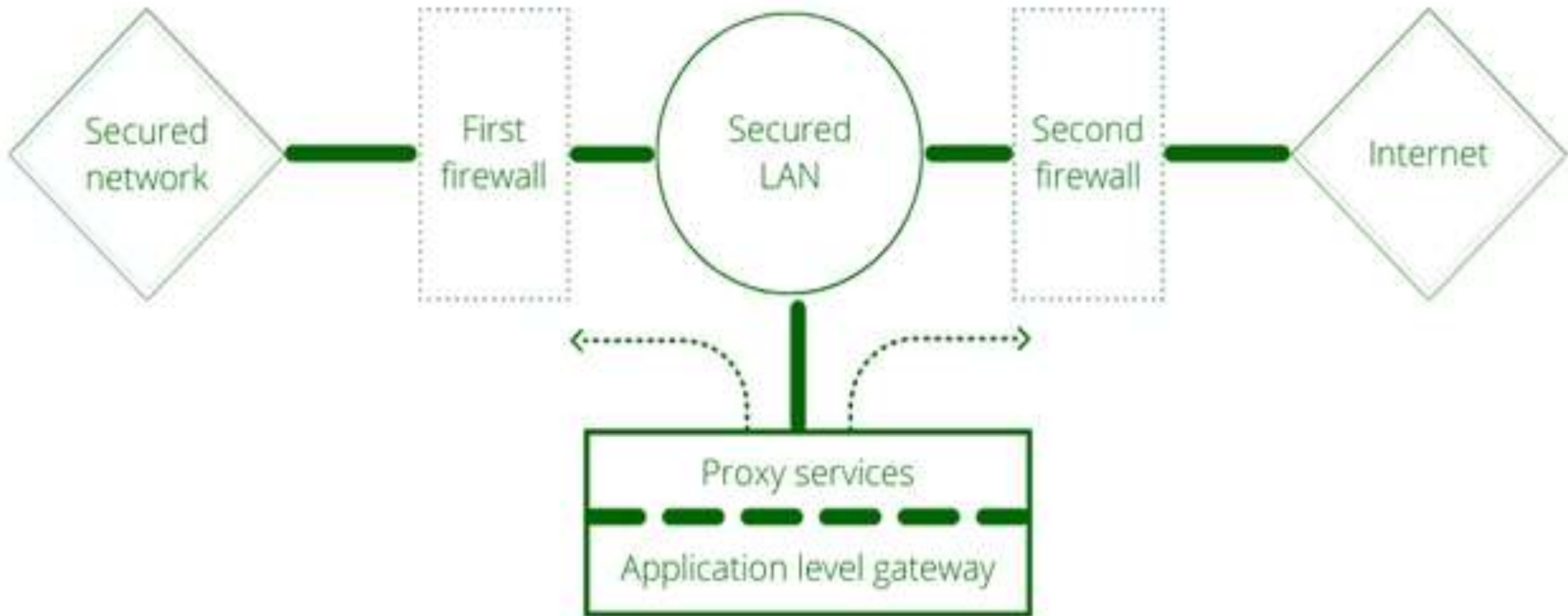
Circuit-Level Gateway

Traffic is filtered based
specific session rules



- Circuit-level gateways monitor TCP handshakes and other network protocol session initiation messages across the network as they are established between the local and remote hosts to determine whether the session being initiated is legitimate, whether the remote system is considered trusted. They don't inspect the packets themselves. However, they provide a quick way to [identify malicious content](#).

Application Gateways –



- also known as a proxy or a proxy firewall, combines some of the attributes of packet filtering firewalls with those of circuit-level gateways. They filter packets according to the service they are intended for (specified by the destination port) and certain other characteristics, such as the HTTP request string.

Next-generation firewall (NGFW)

- [NGFW](#) combines packet inspection with stateful inspection, including a variety of deep packet inspection, along with other network security systems, such as intrusion detection/prevention, malware filtering, and antivirus.

Advantages of using Firewall

- **Protection from unauthorized access:** Firewalls can be set up to restrict incoming traffic from particular IP addresses or networks, preventing hackers or other malicious actors from easily accessing a network or system. Protection from unwanted access.
- **Prevention of malware and other threats:** Malware and other threat prevention: Firewalls can be set up to block traffic linked to known malware or other security concerns, assisting in the defense against these kinds of attacks.
- **Control of network access:** By limiting access to specified individuals or groups for particular servers or applications, firewalls can be used to restrict access to particular network resources or services.

- **Monitoring of network activity:** Firewalls can be set up to record and keep track of all network activity. This information is essential for identifying and looking into security problems and other kinds of shady behavior.
- **Regulation compliance:** Many industries are bound by rules that demand the usage of firewalls or other security measures. Organizations can comply with these rules and prevent any fines or penalties by using a firewall.
- **Network segmentation:** By using firewalls to split up a bigger network into smaller subnets, the attack surface is reduced and the security level is raised.

Disadvantages of using Firewall

- **Complexity:** Setting up and keeping up a firewall can be time-consuming and difficult, especially for bigger networks or companies with a wide variety of users and devices.
- **Limited Visibility:** Firewalls may not be able to identify or stop security risks that operate at other levels, such as the application or endpoint level, because they can only observe and manage traffic at the network level.
- **False sense of security:** Some businesses may place an excessive amount of reliance on their firewall and disregard other crucial security measures like endpoint security or intrusion detection systems.
- **Limited adaptability:** Because firewalls are frequently rule-based, they might not be able to respond to fresh security threats.

- **Performance impact:** Network performance can be significantly impacted by firewalls, particularly if they are set up to analyze or manage a lot of traffic.
- **Limited scalability:** Because firewalls are only able to secure one network, businesses that have several networks must deploy many firewalls, which can be expensive.
- **Limited VPN support:** Some firewalls might not allow complex VPN features like split tunneling, which could restrict the experience of a remote worker.
- **Cost:** Purchasing many devices or add-on features for a firewall system can be expensive, especially for businesses.

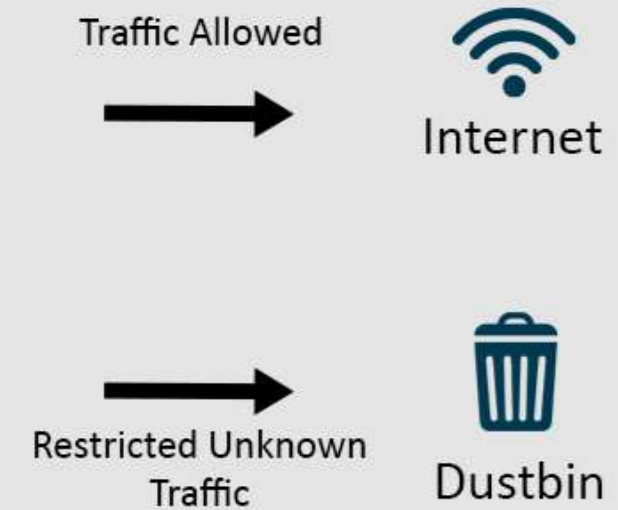
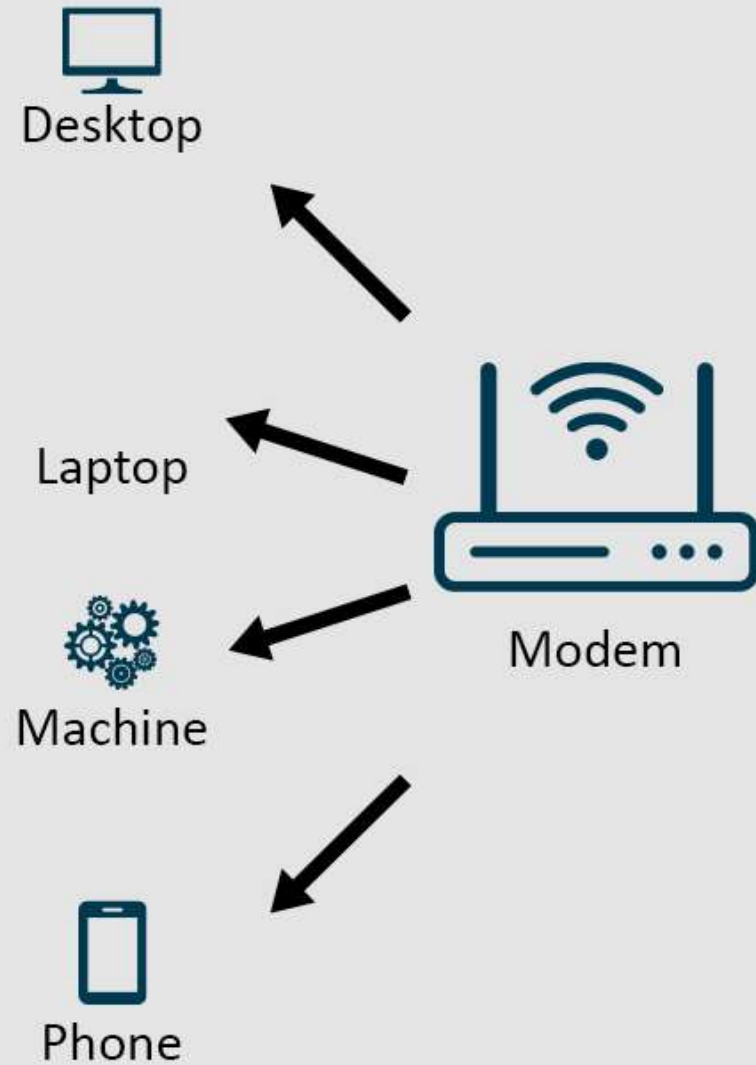
Limitations of a Firewall

- Firewalls are not able to stop the users from accessing the data or information from malicious websites, making them vulnerable to internal threats or attacks.
- It is not able to protect against the transfer of virus-infected files or software if security rules are misconfigured, against non-technical security risks (social engineering)
- It does not prevent misuse of passwords and attackers with modems from dialing in to or out of the internal network.
- Already infected systems are not secured by Firewalls.

Real-Time Applications of Firewall

- **Corporate networks:** Many businesses employ firewalls to guard against unwanted access and other security risks on their corporate networks. These firewalls can be set up to only permit authorized users to access particular resources or services and to prevent traffic from particular IP addresses or networks.
- **Government organizations:** Government organizations frequently employ firewalls to safeguard sensitive data and to adhere to rules like HIPAA or PCI-DSS. They might make use of cutting-edge firewalls like Next-generation firewalls (NGFW), which can detect and stop intrusions as well as manage access to particular data and apps.
- **Service providers:** Firewalls are used by service providers to safeguard their networks and the data of their clients, including ISPs, cloud service providers, and hosting firms. They might make use of firewalls that accommodate enormous volumes of traffic and support advanced features such as VPN and load balancing.

- **Small enterprises:** Small firms may use firewalls to separate their internal networks, restrict access to specific resources or applications, and defend their networks from external threats.
- **Networks at home:** To guard against unwanted access and other security risks, many home users employ firewalls. A firewall that many routers have built in can be set up to block incoming traffic and restrict access to the network.
- **Industrial Control Systems (ICS):** Firewalls are used to safeguard industrial control systems against illegal access and cyberattacks in many vital infrastructures, including power plants, water treatment facilities, and transportation systems.



Secure Private LAN

Public Network

- The firewall operation can be comprehended by considering a simple analogy, where 'IP addresses' are treated as 'houses', and 'port numbers' as 'rooms' within the house. In such a scenario, only trusted people (source addresses) are allowed to get into the house (destination address) at all times. The movement of these people within the house is further filtered or restricted as disclosed below:
- a) The people entering the house are only given access to certain rooms (destination ports), depending on whether they're the owner or a guest.
- b) The owner is allowed to venture into any room (any port), while guests are allowed into a particular set of rooms (specific ports).
- Here, the rules for the kind of activity allowed for an entire bunch of people are pre-defined as and when the firewall is configured on a system or a network. Hence, technically the entire firewall functionality relies on the [monitoring job](#) and allows or blocks the packets based on a set of security protocols.

Computer Intrusions

- When someone tries to access any part of our personal computer system then PC intrusion occurs. Every Personal Computer (PC) which is connected to the internet is a target of hackers and cybercriminals.

Intruder gain access to your computer by:

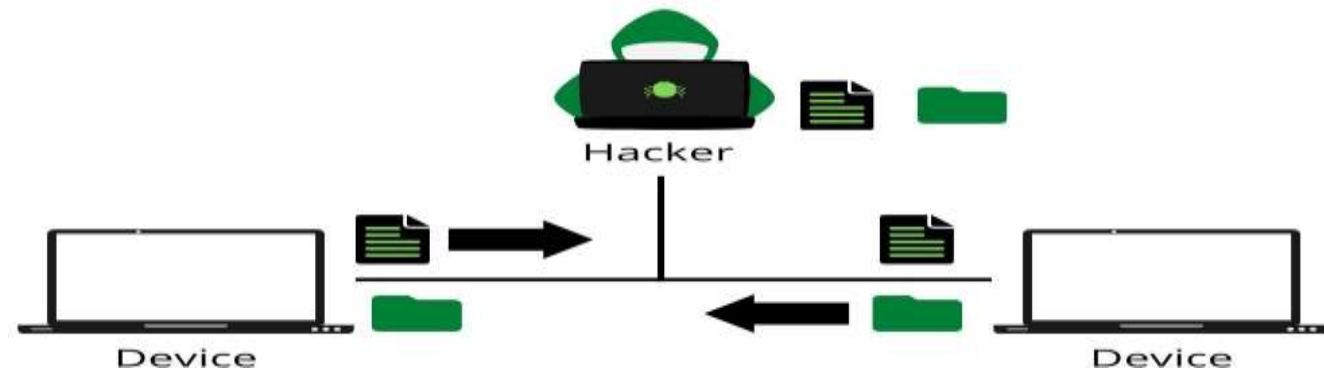
- **Access your computer to view, change, or delete information from your computer:** Once the attacker got access to the computer, initially attacker will attempt to view the information, if the information is valuable attacker may sell the collection information to gain financial benefits. An attacker may delete or change information if the objective of the attack is to distract the smooth functionality of computer.
- **Crash or slow down your computer:** The system may get crash if the attacker deletes system files, bootstrap loader file, bootstrap loader file is responsible to load the operating system. System performance may get slow if the system file is deleted or modified.

- **Access your private data by examining the files on your system:**

Once the attacker got access to the system, he may access private or sensitive data of the system. After analysis of this data, an attacker may get valuable information that may be misused or sold to a third party for financial gain.

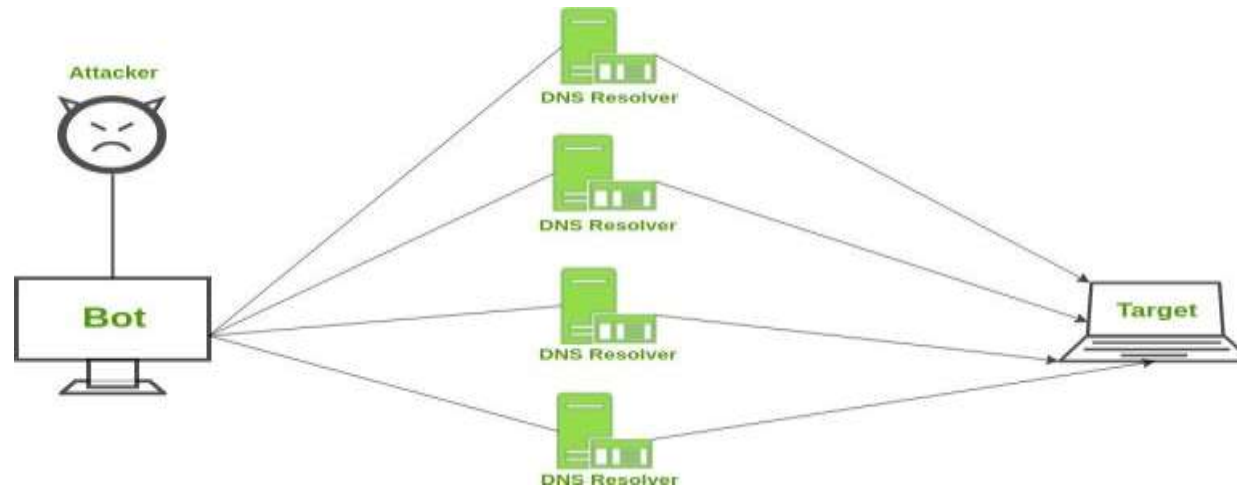
- **Use your computer to access other computers on the Internet**

User's computer is connected in network, if the attacker got access user's system. He may use the user machine to get access to another network machine by executing various commands such as ping, traceroute, dig, etc.

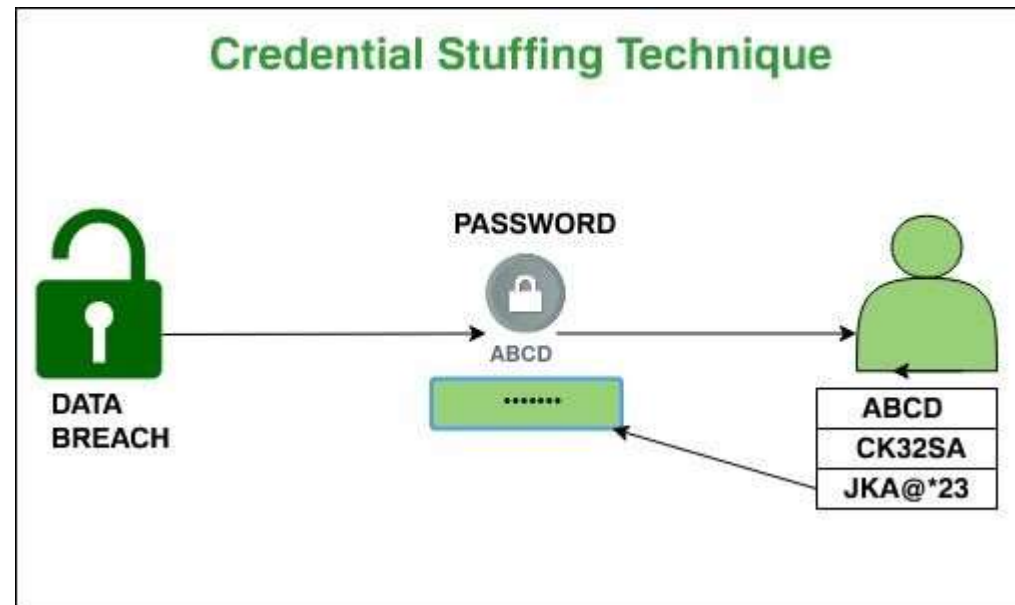


Personal Computer Intrusion can occur in any of the following forms:

- **Sweeper Attack:** Cybercriminals erase all the information or data like cache, cookies, internet history, or documents from the system by a malicious program.
- **Denial of Services:** [DDos](#) type of attack in which attackers may shut down the PC services making it inaccessibile to its original user. All the system applications and stored resources come to a halt.



- **Password Guessing:** Most hackers crack passwords of system accounts by guessing and gaining remote entry into our personal computer system. Hackers can use this form and may damage the security system in our PC.



- **Snooping:** Snooping refers to opening and looking through files in an unauthorized manner. Snooping may involve many types of things such as gaining access to data in an unauthorized way, casually observing someone else's email, or monitoring the activity of someone else's computer through sophisticated snooping software. It involves monitoring keystrokes pressed on the keyboard, capturing of passwords and login information, interception of emails and other private communication, and data transmission.
- **Eavesdropping:** When cyber vandals or attackers listen to a conversation that is traveling over devices like computers, servers, or other network devices, it is called eavesdropping. Formally we can say that the intentional interception of someone else's data as it passes through a user's computer to a server or vice-versa is called [eavesdropping](#).

What is Network Intrusion?

- A network intrusion is any illegal activity carried out on a digital network.
- Network incursions frequently entail the theft of valuable network resources and virtually always compromise a network security and/or data security.
- Organizations and their cybersecurity teams must have a comprehensive understanding of how network intrusions operate and implement network intrusion, detection, and response systems that are designed with attack techniques and cover-up methods in mind
- in order to detect and respond proactively to network intrusions.

A network is often compromised for one of the following three reasons:

- ***Hacktivism***: Hacktivism is the combination of hacking with activism. Intruders that wish to hack to prove a political agenda or social cause are responsible.
- ***Steal Money***: This infiltration aims to steal money or data from the victim. Typically, the intent is to abuse the other person for monetary advantage.
- ***Spying***: Spying is the entrance of a state-sponsored network in order to spy on an adversary.

- Network intrusion attacks can originate from individuals, major corporations, or even governments.
- These organizations' cybersecurity teams must comprehend how network intrusion is carried out in order to properly prevent it.
- A Network Intrusion Detection System must be implemented in order to address network intrusion-related difficulties.
- There are two sorts of systems that can aid in the prevention of network attacks: **intrusion detection and prevention systems**.
- **Intrusion Detection System (IDS)** is a passive system that detects harmful behavior on a network,
- **Intrusion Prevention System** not only detects but also blocks the same dangerous activity.

What are the Risks of Network Intrusion?

- stealing personal information.
- to obtain access to the internal systems.
- Network intrusions include DDoS (distributed denial of service), SQL injection, and Man in the Middle(MitM), among others.

major risks of network intrusion

- **Corruption of Data:**

- organization's or customers' vital data.

- The status of orders and workflows may shift, and client payments may become delayed. During audits, ledgers and tainted financial [data](#)

- **Financial Loss for the Organization:**

- to coordinate with third-party organizations that will handle and mitigate the attack on their behalf. It is also likely that the organization gets taxed based on the number of requests, which will only make matters worse.

- **Theft of Data:**

- personal information of consumers.
- access to cameras and contacts might pose far greater risks to their consumers.

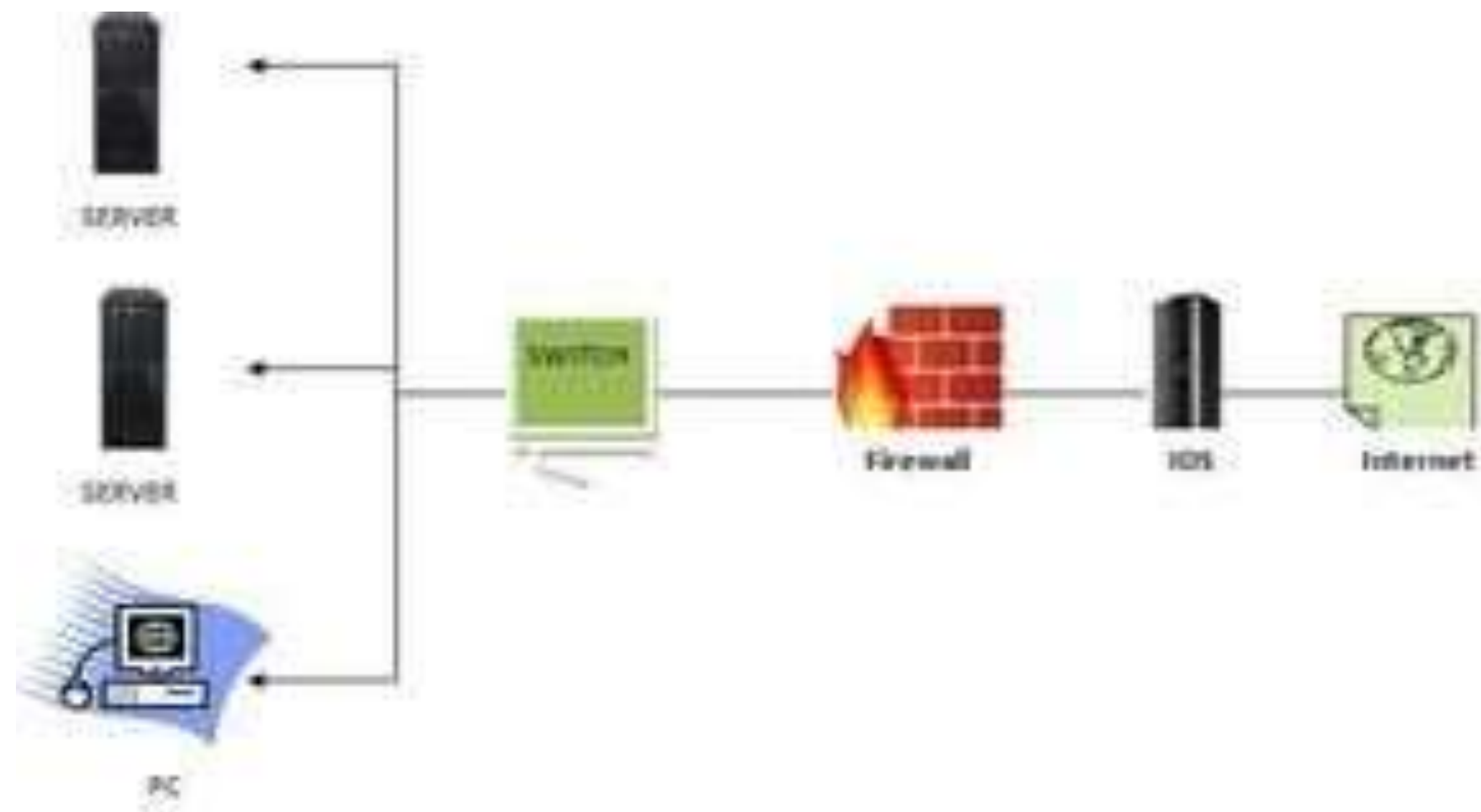
- **Operational Disruption:**

- delay in the workflows of operations.

- **Loss of Reputation:** Reputation loss may be disastrous for a company. Loss of clients.

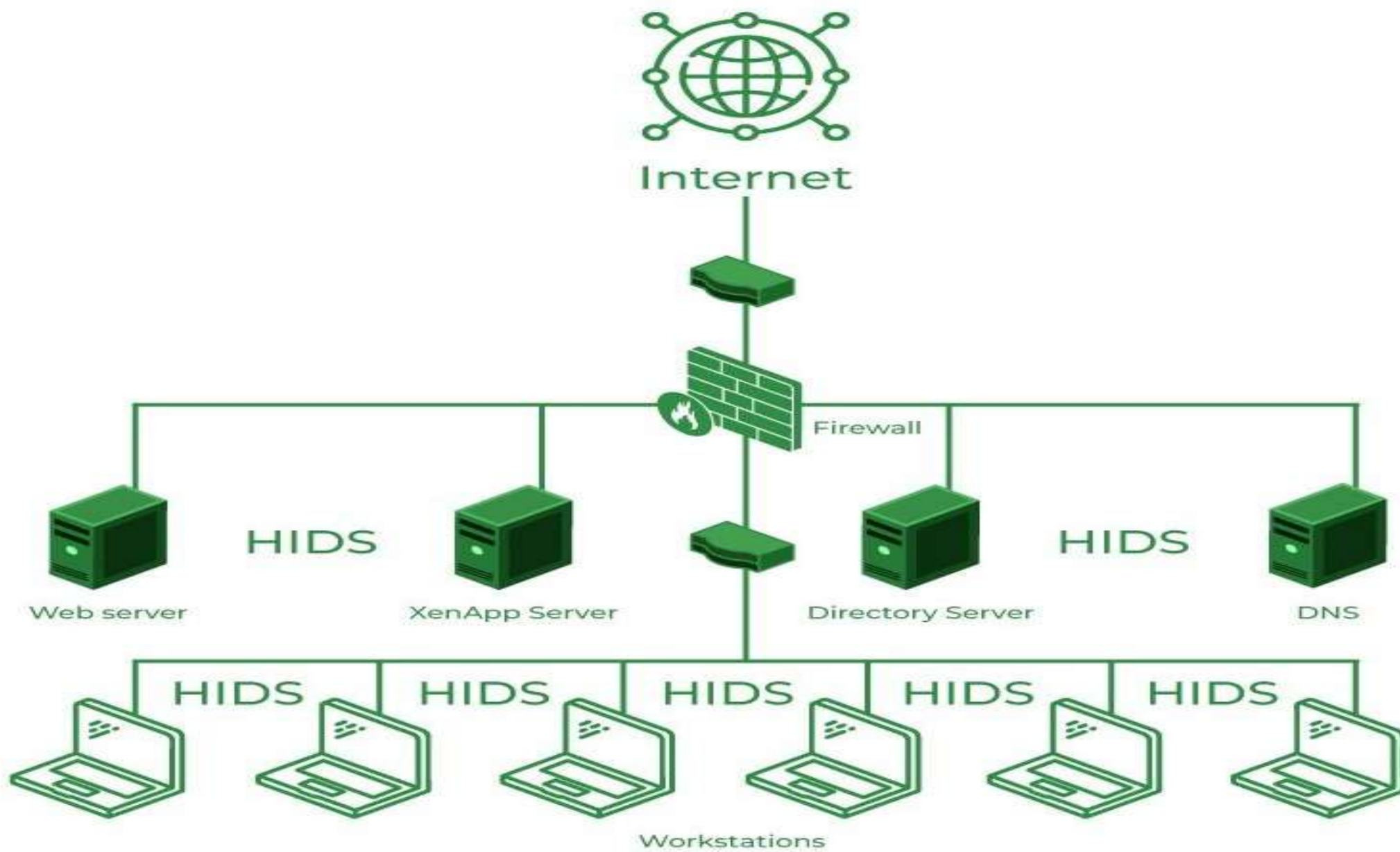
How to Detect Network Intrusion?

- intrusion detection system (IDS) observes network traffic for malicious transactions and sends immediate alerts when it is observed.
- It is software that checks a network or system for malicious activities or policy violations.
- Each illegal activity is recorded either or notified to an administration.
- IDS monitors a network or system for malicious activity and protects a computer network from unauthorized access from users, including perhaps insiders.
- to build a predictive model (i.e. a classifier) capable of distinguishing between 'bad connections' (intrusion/attacks) and 'good (normal) connections'.



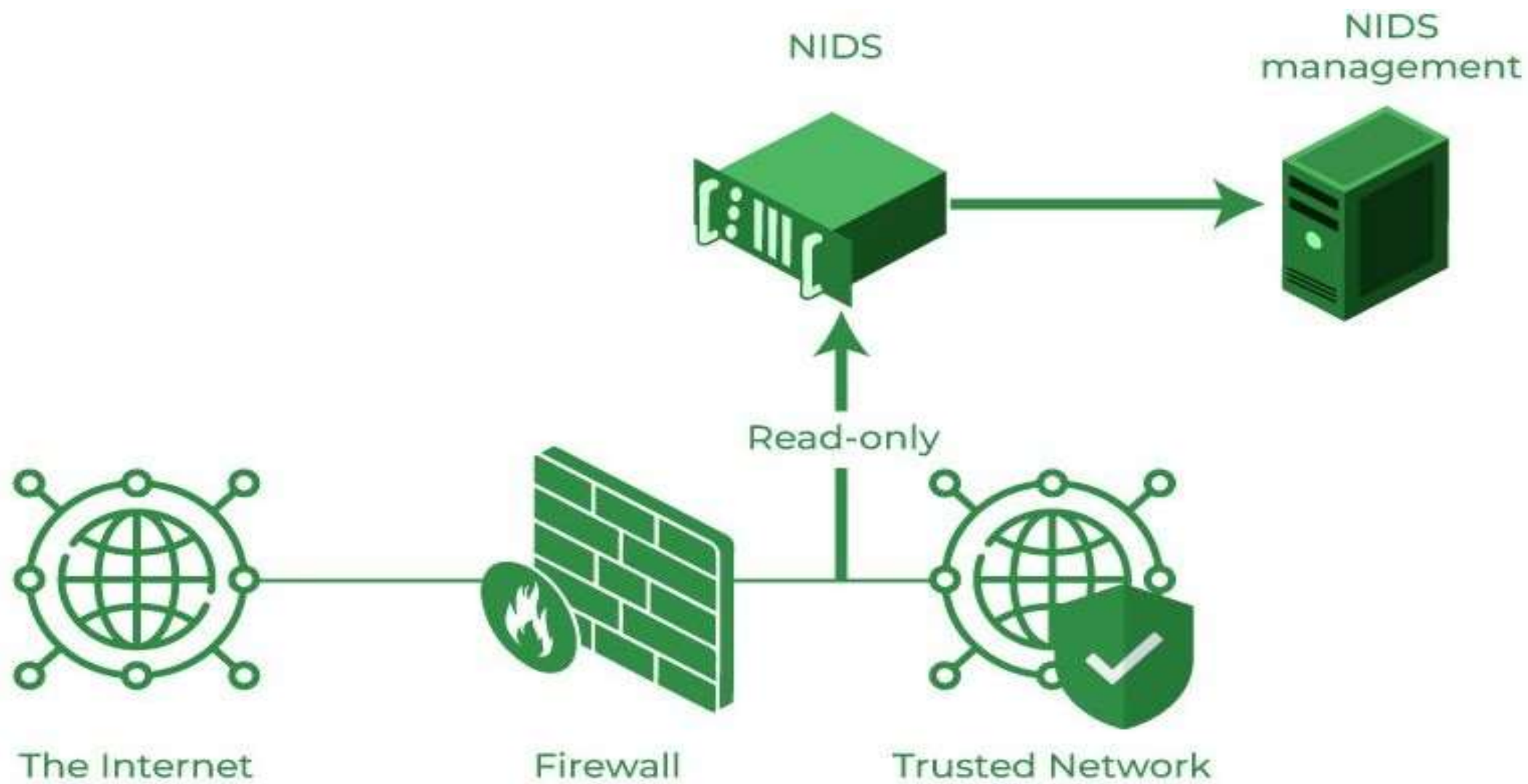
Classification of Intrusion Detection System(IDS)

- **Host Intrusion Detection System:** This system operates on network hosts or standalone devices.
- monitors the incoming and outgoing packets from the device only and will alert the administrator if suspicious or malicious activity is detected.
- It takes a snapshot of the current system files and compares it to previous snapshots.
- Similarly, if the analytical system files are modified or destroyed, an alert is sent to the administrator for further inquiry.



- **Network Intrusion Detection System:**

- located across the network to monitor traffic from all devices connected to the Internet.
- it performs an analysis of passing traffic on the whole subnet and compares that information to a database of known threats.
- When it detects any strange activity, it alerts the administrator.
- It is accomplished largely through the following methods:
 - Monitoring system setups and settings
 - Monitoring user behavior in order to identify nefarious intent
 - Scanning techniques that discover hazardous pattern indicators
 - System file comparisons against malware signatures



- **Protocol-based Intrusion Detection System (PIDS):**
- It attempts to protect the web server by continuously checking the HTTPS protocol stream and allowing the associated HTTP protocol.
- As HTTPS is not secured, this system would need to live in this interface prior to accessing its web presentation layer in order to use HTTPS.

- **Application Protocol-based Intrusion Detection System (APIDS):**

- agent that often lives within a cluster of computers.
- It discovers intrusions by monitoring and analyzing application-specific protocol traffic.
- This would monitor, for instance, the SQL protocol specific to the middleware as it interacts with the database on the web server.

- **Hybrid Intrusion Detection System:**

- two or more intrusion detection system techniques.
- host agent or system data is coupled with network data to get a comprehensive view of the network system.
- The effectiveness of the hybrid intrusion detection system surpasses that of the other intrusion detection.

How to Prevent Network Intrusion?

- Typically, intrusion prevention systems are placed behind a [firewall](#) to serve as an additional filter against malicious activities.
- As a result of their in-line placement, intrusion prevention systems are able to analyze and automatically respond to all network traffic flows.
- These steps may include informing administrators, discarding harmful packets, suspending traffic from the malicious activity's originating address(es), and resuming connections.
- Importantly, a good intrusion prevention system must be efficient so as not to degrade network performance.
- In order to detect harmful activity in real-time and minimize false positives, intrusion prevention systems must also function swiftly and correctly.

What is a Network Intrusion Prevention System?

- Intrusion prevention systems are a kind of [network security devices](#) that monitor for harmful network or system activity.
- Indeed, the primary duties of an intrusion prevention system (IPS) are to recognize harmful behavior, collect information about it, report it, and attempt to block it.
- Both Intrusion Prevention Systems and Intrusion Detection Systems monitor network traffic and system activity for harmful behavior.
- [IPS](#) can take proactive measures, such as sending an alert, resetting a connection, or blocking traffic from a malicious IP address.

- There are four subtypes of intrusion prevention systems, each of which is briefly discussed below.
- **Network Behavior Analysis:** Network Behavior Analysis([NBA](#)) system meticulously monitors network traffic in order to discover dangers that create abnormal traffic flows, such as DDoS attacks, a sort of malware.
- **Network-based Intrusion Prevention System(NIPS):** [NIPS](#) searches the whole network for protocol analysis of unusual traffic.
- **Host-based Intrusion Prevention Systems:** They are the software packages deployed and configured to monitor a single host for suspicious behavior by identifying the host's internal events.
- **Wireless Intrusion Prevention Systems:** Examining the wireless networking standards, it analyzes wireless networks for any suspicious activities.

Detection Method of IDS

Signature-based Method: patterns such as the number of bytes or a number of 1s or the number of 0s in the network traffic.

It also detects on known malicious instruction sequence that is used by the malware.

Anomaly-based Method: Anomaly-based IDS was introduced to detect unknown malware attacks as new malware is developed rapidly.

The machine learning-based method has a better-generalized property in comparison to signature-based IDS as these models can be trained according to the applications and hardware configurations.

Trusted Systems in Network Security

- A trusted system is a computer system or network that has been designed, implemented, and tested to meet specific security requirements.
- to protect sensitive information, prevent unauthorized access, and ensure the integrity and availability of data and systems.

Trusted Systems are based on different level of security.

- **Multilevel Security:** This type of Trusted system ensures that security is maintained at different levels of the computer system. It ensures that the information is prevented from being at risk. The different security levels of computer systems are :
 - Top Secret Level
 - Secret Level
 - Confidential Level
 - Unclassified Level
- The order of security level is also given by top level security having the highest priority followed by secret Level priority, confidential Level priority and then least priority is assigned to unclassified level priority. If security is not cleared at one particular level, flow of information is restricted. Also, one important point that must be kept in mind is that 'Read Up' and 'Write Down' are not permitted in multilevel security.

- **Data Access Control:** This type of Trusted system provides additional security to the verified process of log-in. It helps in setting permissions for different users, giving them limited access and restricting any additional accesses granted. There are three basic models of Data Access Control:
 - **Access Matrix:** They are composed of three parts
 - Subject
 - Object
 - Access right
 - **Access Control List:** They are composed of different entries of objects depicting user access and the level of access granted (public or private). Access control list demonstrate column-wise split.
 - **Capability List:** They are composed of authorized users and the granted operations for them. Users can have multiple capability tickets. Capability list demonstrate row-wise split.

- **Reference Monitor:**

- This type of trusted system provides hardware level security by limiting the access to objects.
- Reference monitor maintain security rules ensuring that 'Read Up' and 'Write Down' operations are not performed.
- Reference monitor ensure that the entire security maintaining process that is carried out is verified and safe.

Importance of Trusted System:

- **Identity Verification:** Trusted systems ensure that only verified users are given access. The verification process takes place that each user is identified uniquely.
- **Safety Maintained:** Trusted system ensures that safety is maintained by preventing direct access to confidential information.
- **Limiting Access:** Permissions and access that are absolutely necessary are granted for users. Unwanted rules and permissions are avoided.
- **Preventing Malicious Activities:** Trusted systems have a mechanism in place to detect and prevent malicious activities such as hacking attempts and unauthorized access.
- **Ensuring Compliance:** Trusted systems help organizations to comply with various regulations and standards such as HIPAA, PCI-DSS, and SOX by providing a secure environment for sensitive information.

These principles of TS:

Least Privilege:

minimum level of access necessary to perform their tasks.

Defense in Depth:

This principle involves using a combination of physical, technical, and administrative controls.

Integrity:

Trusted systems ensures

Confidentiality:

Trusted systems protect sensitive information from unauthorized access.

Availability:

This principle ensures that critical information and systems are accessible and operational at all times.

TS use technologies and techniques

Hardware-based security: hardware, such as secure processors, to provide a secure environment for critical operations.

Virtualization: Virtualization is a technique that is often used in trusted systems to create multiple virtual machines running on a single physical machine.

Multi-factor authentication: Trusted systems often use multi-factor authentication to verify the identity of users.

Encryption: to protect sensitive data.

Auditing: use auditing to track and monitor system activity.

Password Management

- It is a series of letters or numbers that you must type into a computer or computer system in order to be able to use it.
- A password is a **real-life implementation of challenge-response authentication.**

Password Management vulnerabilities:

- two ways:

1. From end user:

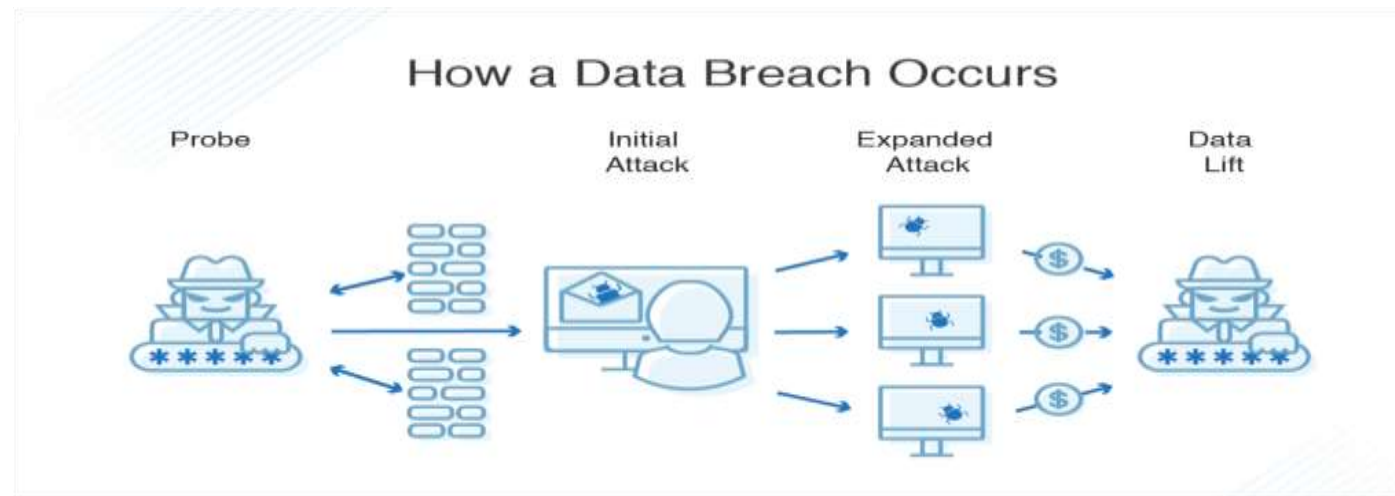
- Weak and easy to guess.(DOB/Name)
- Rarely changed.(Change password)
- Same password to all sites.
- Noting password in non secure place.

2. Technical Vulnerabilities:

- Weak encryption technique
- Application that displaying password on screen while typing

Issues Related to Managing Passwords:

- Login spoofing Ex: email that looks like it comes from PayPal
- Sniffing attack Ex: Attackers can capture login credentials
- [Brute force attack](#) Ex: to logically guess your credentials
- [Shoulder surfing attack](#)
- Data breach



Methods to Manage Password:

- **Strong and long passwords:**
- 8 to 12 characters long, also at least three different character sets
- **Password Encryption:** Using end-to-end encryption is recommended.
- **Multi-factor Authentication (MFA):**
- Adding some security questions ,phone number
- **Make the password pass the test:** it falls under the strong and safe password category.
- **Avoid updating passwords frequently:**
- frequently as in 60 or 90 days.

Password Selection/Protection Strategies:

1. User education Ex: Training...Security
2. Using computer generated password Ex: Very hard to predict..
3. Reactive password checking Ex: Periodical checking for Authentication
4. Pro-active password checking Ex: Pre-conditions

Pros Of Password Managers

1. **Ease Of Use:** Password managers are easy to use. Passwords are **easily accessible** and **quick to load**.
 2. **Mobile Compatible :** Password managers can even **work on mobile devices**
 3. **Enhanced Security :** Password managers add **another layer of protection** for users.
it is **more likely** that they will use **unique, complex passwords** for each application.
- And, **some password managers will even generate secure passwords for you!**

Cons Of Password Managers

1. Single Point Of Failure

- **an attacker could gain access to all of your passwords with one hack of your password manager.**

2. Stored In The Cloud

- **Not everyone has implemented cloud technology** in their workplace.

3. Extra Step To Login

- Although password managers simplify the login procedure, the **perception is that it adds a step to the login procedure**. With the focus on speed of access

Popular password managers

From sources across the web



1Password



LastPass



RoboForm



ASH

Dashlane



KeePass



MeO
passwords & Smarter

LogMeOnce Password M...



Bitwarden



NordPass



Norton Password Manager



